

CIS Research in Constructive Analysis

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Lecture 1: Proof of the problems in syllabus

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Lemma 1 (Equality test not computable). The equality relation $=$ on $\mathbb{R}$ is not decidable.

Proof. Suppose there exists a representation $\delta : \subseteq \Sigma^\omega \rightarrow \mathbb{R}$ where the equality relation is δ -decidable. Consider a real number x and some p such that $\delta(p) = x$. On input $\langle p, p \rangle$, the Turing machine will conclude that the inputs are equal, after a finite number of steps. Let w be the prefix of p that the machine has read before it stops. We know $w \in \Sigma^*$, i.e. is a finite string. Now for any other real number x' with $\delta(p') = x'$, the machine will conclude that $x = x'$ whenever $p' \in w\Sigma^\omega$, i.e. p' has w as a prefix.

Let $\sim$ be the equivalence relation on Σ^ω generated by the machine, where $p \sim p' \Leftrightarrow$ the machine concludes after a finite number of steps that $\delta(p) = \delta(p')$. The equivalence classes $\Sigma^\omega / \sim$ are determined by finite strings in Σ^* , however Σ^* is countable, so $\Sigma^\omega / \sim$ is countable. Since δ is onto, we conclude that $\mathbb{R}$ is countable, a contradiction. $\square$

Lemma 2. Comparison with 0 is not computable for reals.

Proof. $x = y \Leftrightarrow x - y = 0$, so this is equivalent to saying that equality test is decidable, which is false by the last lemma. $\square$

Lemma 3. Comparison with 0 is not computable for constructive reals.

Proof. This one I do not know how to prove for general δ . Take δ to be the Cauchy representation. Take a sequence of rational numbers that converge rapidly to 0, i.e. $|x_i| < 2^{-i}$. Denote the terms by $x_1, x_2, \dots$. This sequence generates a Cauchy representation of 0. Suppose the comparison is computable, then the Turing machine will stop after reading finite n terms from the Cauchy sequence, and conclude that the number it represents is exactly 0.

Now any other constructive real $y \in \bigcap_{i=1}^n (x_i - 2^{-i}, x_i + 2^{-i})$ can be represented by the same $x_1, \dots, x_n$, but not necessarily equal to 0. We know such a y exists because $\mathbb{R}_c$ is dense in $\mathbb{R}$. $\square$

Theorem 1 (problems in syllabus). Consider a linear equation $Ax + By + C = 0$ where $A, B, C \in \mathbb{R}_c$. There is no algorithm that tells if the equation has a root, if it has exactly one root, or on which endpoint of a closed interval of x it attains maximum for y .

Proof. The linear equation has no root iff $A = B = 0$ and $C \neq 0$. It has infinitely many roots iff $A = B = C = 0$. It has exactly one root iff either $A \neq 0$ or $B \neq 0$. The RHS of the equivalences above are simply comparison with 0, which are not computable by the last lemma.

If we restrict the domain to $\{(x, y) \in \mathbb{R}_c^2 : a \leq x \leq b\}$, it can be easily shown that the endpoint on which y attains maximum only depends on sign of the gradient $\frac{dy}{dx} = -\frac{A}{B}$. Under the Cauchy representation, the arithmetic operations are computable, hence $-\frac{A}{B}$ is computable. The problem then reduces to a comparison with zero, which is not computable by the last lemma. $\square$